Early Detection of Overfitting via Persistent Homology Analysis of Deep Neural Network Activations

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Abstract

Overfitting constitutes a fundamental challenge in deep learning. Classical detection methods rely on monitoring validation loss, often a lagging indicator. In this work, we propose an approach based on Topological Data Analysis (TDA) to detect early signs of overfitting. We compute the persistent homology of internal activations of neural networks trained under controlled overfitting conditions. Our study covers three datasets (Fashion-MNIST, CIFAR-10, MNIST), three architectures (shallow CNN, deep CNN, MLP), and four training set sizes, following a one-factor-at-a-time design (8 configurations total), with five independent runs per configuration to ensure statistical reproducibility. The topological invariants—in particular the smoothed derivative of $\mathrm{PE}(H_1)$ and β_1—are quantitatively compared to four classical detectors (early stopping, loss gap, gradient norm, weight norm). Our results show that detection based on the derivative of $\mathrm{PE}(H_1)$ triggers on average at epoch 23.6 ± 6.1 on Fashion-MNIST/CNN, compared to 77.0 for early stopping (patience = 10), while maintaining a 100% detection rate across all tested configurations. The stability of topological invariants with respect to subsampling is validated through bootstrap analysis.

Keywords: Topological data analysis, persistent homology, overfitting, deep neural networks, persistent entropy, Betti numbers, change-point detection.

1 Introduction

Deep learning has revolutionized numerous fields, from computer vision to natural language processing. However, a recurrent problem persists: overfitting, where the model memorizes training data at the expense of its generalization capacity [1].

Classical methods for detecting overfitting primarily rely on monitoring validation loss (early stopping) or indirect indicators such as weight norms, gradient norms, or measures of loss surface "flatness" [2]. These approaches have limitations: early stopping is a lagging indicator that only signals degradation after it has set in; weight and gradient norms are indirect proxies that do not characterize the geometry of internal representations.

Topological Data Analysis (TDA) offers a rigorous mathematical framework for characterizing the "shape" of data across scales [3, 4]. In particular, persistent homology enables the detection of topological structures (connected components, loops, cavities) and tracks their evolution throughout a filtration [5, 6]. Recent work has shown that the topology of internal representations in a neural network evolves during training [1, 7, 8]. Naitzat et al. [1] demonstrated that successive layers of a deep network progressively simplify data topology. Rieck et al. [7] proposed neural persistence as a complexity measure. Corneanu et al. [9] explored the links between decision boundary topology and generalization error.

Building on these observations, an important question is whether the evolution of internal representation topology can serve as an early indicator of overfitting, and how it compares to classical detection methods.

Contributions. Based on these questions, this work provides the following contributions:

1. A systematic study of topological evolution of activations under controlled overfitting, covering 3 datasets, 3 architectures, and 4 training set sizes in a one-factor-at-a-time design (8 configurations total).
2. A quantitative comparison between topological detectors (CUSUM and smoothed derivative on $\mathrm{PE}(H_1)$ and β_1) and four classical detectors (early stopping, loss gap, gradient norm, weight norm), with 5 independent runs per configuration.
3. A stability analysis via bootstrap of topological invariants with respect to subsampling.
4. The identification of a three-phase pattern (compression, stabilization, fragmentation), where the third phase constitutes a potential precursor signal of overfitting.

2 Mathematical Foundations

We briefly recall the essential concepts. For a complete introduction, see [10, 4].

2.1 Simplicial Complexes and Vietoris-Rips Filtration

Definition 2.1 (Vietoris-Rips Complex)**.** *Let $X = \{x_1, \ldots, x_N\} \subset \mathbb{R}^D$ be a point cloud and $\varepsilon \geqslant 0$. The Vietoris-Rips complex $\mathrm{VR}_\varepsilon(X)$ is the simplicial complex:*

$$\mathrm{VR}_\varepsilon(X) = \big\{\sigma \subseteq X \;\big|\; d(x_i, x_j) \leqslant \varepsilon \;\; \forall\, x_i, x_j \in \sigma\big\}.$$

By increasing ε from 0 to $+\infty$, we obtain a filtration $\emptyset = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m$.

2.2 Persistent Homology and Betti Numbers

Homology associates to each complex K abelian groups $H_k(K)$, $k \in \mathbb{N}$. The k-th Betti number is:

$$\beta_k(K) = \mathrm{rank}\big(H_k(K)\big). \tag{1}$$

Intuitively, β_0 counts connected components and β_1 counts independent loops. In a filtration, each topological feature γ has a birth time $b(\gamma)$ and a death time $d(\gamma)$, forming the persistence diagram $\mathrm{Dgm}_k = \{(b_i, d_i)\}$ [5].

2.3 Persistent Entropy

Definition 2.2 (Persistent Entropy [11])**.** *Let* $\mathrm{Dgm} = \{(b_i, d_i)\}_{i=1}^n$ *be a persistence diagram (finite features). Define* $\ell_i = d_i - b_i$ *and* $L = \sum_{i=1}^n \ell_i$*. The persistent entropy is:*

$$\mathrm{PE}(\mathrm{Dgm}) = -\sum_{i=1}^{n} \frac{\ell_i}{L} \ln\left(\frac{\ell_i}{L}\right). \tag{2}$$

PE *reaches its maximum* $\ln(n)$ *when all lifetimes are equal (maximal complexity) and equals* 0 *when a single feature dominates. This measure is stable with respect to diagram perturbations [11].*

3 Methodology

3.1 Datasets

We use three image classification datasets:

- Fashion-MNIST [12]: 70,000 grayscale 28×28 images, 10 clothing classes.
- MNIST [13]: 70,000 grayscale 28×28 images of handwritten digits, 10 classes.
- CIFAR-10 [14]: 60,000 color 32×32 images (3 channels), 10 object classes.

To induce controlled overfitting, the training set is restricted to a randomly selected subset of size n_{train}, while the complete test set is retained for evaluation. We test four sizes: $n_{\text{train}} \in \{200, 500, 1000, 2000\}$.

3.2 Architectures

Three architectures are compared, all equipped with a *bottleneck* of dimension $D = 32$ which activations are the object of topological analysis:

1. CNN Shallow: 2 convolutional blocks [Conv3×3–BN–ReLU–MaxPool]×2, adaptive pooling, dense layer $64 \to 32$ (bottleneck), classifier $32 \to 10$.
2. CNN Deep: 4 convolutional blocks ($32 \to 64 \to 128 \to 128$ filters), adaptive pooling, bottleneck $128 \to 32$, classifier.
3. MLP: two dense layers ($d_{\text{in}} \to 256 \to 128$), bottleneck $128 \to 32$, classifier. The image is flattened at input.

3.3 Training Protocol

Each configuration is trained for $T = 80$ epochs with the Adam optimizer [15] ($\eta = 10^{-3}$), cross-entropy loss, and no regularization. The experimental design follows a one-factor-at-a-time (OFAT) protocol with F-MNIST / Shallow / $n = 500$ as the reference configuration: dataset, architecture, and training size are each varied independently while the other two factors are held fixed, yielding 8 distinct configurations. Each is executed 5 times with different random seeds ($s \in \{42, 123, 456, 789, 1024\}$), for a total of $8 \times 5 = 40$ training runs.

At each epoch, we record in addition to classical metrics (loss, accuracy):

- The L^2 *gradient norm* $\|\nabla_\theta \mathcal{L}\|_2$ (averaged over batches within the epoch);
- The L^2 *weight norm* $\|\theta\|_2$.

3.4 Topological Analysis Protocol

At the end of each epoch t, $N = 500$ validation examples are propagated, and bottleneck activations are collected via PyTorch hooks, producing $A^{(t)} \in \mathbb{R}^{N \times 32}$. Persistent homology is computed via Ripser [16], producing $\mathrm{Dgm}_0^{(t)}$ and $\mathrm{Dgm}_1^{(t)}$. Algorithm 1 formalizes this protocol.

Algorithm 1: Topological and conventional monitoring protocol.

Input : Network f_θ, $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{val}}$, number of epochs T
Output: Time series $\{\beta_k^{(t)}, \mathrm{PE}_k^{(t)}, \|\nabla\mathcal{L}\|^{(t)}, \|\theta\|^{(t)}\}_{t=1}^{T}$
for $t = 1, \ldots, T$ **do**
  Train f_θ for one epoch; record $\|\nabla\mathcal{L}\|^{(t)}$ and $\|\theta\|^{(t)}$;
  Compute $\mathcal{L}_{\text{train}}^{(t)}$, $\mathcal{L}_{\text{val}}^{(t)}$;
  Extract $A^{(t)} \in \mathbb{R}^{N \times 32}$;
  Compute $\mathrm{Dgm}_0^{(t)}, \mathrm{Dgm}_1^{(t)}$ via Ripser;
  $\beta_k^{(t)} \leftarrow |\{(b,d) \in \mathrm{Dgm}_k^{(t)} : d < +\infty\}|$; $\quad \mathrm{PE}_k^{(t)} \leftarrow -\sum_i \frac{\ell_i}{L} \ln \frac{\ell_i}{L}$;
end

3.5 Compared Overfitting Detectors

We compare six detectors, each returning the detection epoch t^* or $\varnothing$ (not detected):

1. Early stopping (ES): t^* = first epoch where validation loss has not decreased for $p = 10$ consecutive epochs.
2. Loss gap (LG): t^* = first epoch where $\mathcal{L}_{\text{val}} - \mathcal{L}_{\text{train}} > 0.1$ for 5 consecutive epochs.
3. Weight norm (WN): t^* = first epoch where the local slope of $\|\theta\|_2$ (linear regression over a 10-epoch window) is strictly positive.
4. Gradient norm (GN): t^* = first epoch where $\|\nabla\mathcal{L}\|^{(t)} > 2 \cdot \overline{\|\nabla\mathcal{L}\|}$ over a sliding 10-epoch window.
5. TDA – Smoothed derivative (TDA-D): detection of the first sign change (negative $\rightarrow$

positive) in the derivative of the moving average (window 10) of PE(H_1) or β_1, after a warm-up period of 15 epochs.

6. TDA – CUSUM (TDA-C): CUSUM algorithm (Cumulative Sum) applied to PE(H_1) and β_1, with threshold h calibrated separately for each metric.

3.6 Bootstrap Stability Analysis

To assess the sensitivity of topological invariants to subsampling, we perform at key epochs ($t \in \{1, 10, 20, 40, 60, 80\}$) a bootstrap procedure: 10 draws without replacement of 80% of the $N = 500$ activation points, followed by persistent homology computation on each subsample. Intra-bootstrap variance (sensitivity to sampling) is compared to inter-run variance (sensitivity to training stochasticity).

4 Results

The complete set of experiments represents 8 configurations $\times$ 5 runs = 40 training sessions, each of 80 epochs with topological analysis at each epoch.

4.1 Observed Overfitting

Table 1 summarizes the final accuracy gap (train – val) for each configuration, averaged over the 5 runs. Overfitting is systematically observed, with a growing gap as n_{train} decreases.

Table 1: Final accuracy gap (train – val) in percentage points, averaged over 5 runs ($\mu \pm \sigma$).

Dataset	Architecture	n_{train}	ΔAcc (%)
Dataset variation (CNN Shallow, $n = 500$)			
Fashion-MNIST	CNN Shallow	500	16.4 ± 4.0
CIFAR-10	CNN Shallow	500	23.1 ± 1.8
MNIST	CNN Shallow	500	15.1 ± 2.6
Architecture variation (Fashion-MNIST, $n = 500$)			
Fashion-MNIST	CNN Shallow	500	16.4 ± 4.0
Fashion-MNIST	CNN Deep	500	20.4 ± 0.5
Fashion-MNIST	MLP	500	22.7 ± 0.4
Training size variation (Fashion-MNIST, CNN Shallow)			
Fashion-MNIST	CNN Shallow	200	16.3 ± 4.1
Fashion-MNIST	CNN Shallow	500	16.4 ± 4.0
Fashion-MNIST	CNN Shallow	1000	9.9 ± 1.3
Fashion-MNIST	CNN Shallow	2000	11.4 ± 1.3

4.2 Comparison of Detection Epochs

The central result of this study is presented in table 2. For each detection method and each configuration, we report the average detection epoch $\overline{t^*} \pm \sigma$ over the 5 runs, as well as the detection rate (fraction of runs that triggered an alarm before the end of training).

Table 2: Average detection epoch ($\overline{t^*} \pm \sigma$) and detection rate (in parentheses) for each method. ES = early stopping, LG = loss gap, WN = weight norm, GN = gradient norm, TDA-D = smoothed derivative PE(H_1), TDA-D$_\beta$ = smoothed derivative β_1.

Config.		ES	LG	WN	GN	TDA-D PE(H_1)	TDA-D$_\beta$ β_1
F-MNIST / Shallow	n=500	77.0 (1/5)	23.8 ± 21.4 (5/5)	11.0 ± 0.0 (5/5)	— (0/5)	23.6 ± 6.1 (5/5)	21.6 ± 4.7 (5/5)
CIFAR-10 / Shallow	n=500	— (0/5)	20.2 ± 4.2 (5/5)	11.0 ± 0.0 (5/5)	42.6 ± 7.3 (5/5)	18.4 ± 2.2 (5/5)	19.8 ± 3.1 (5/5)
MNIST / Shallow	n=500	— (0/5)	27.6 ± 4.4 (5/5)	11.0 ± 0.0 (5/5)	— (0/5)	27.4 ± 9.3 (5/5)	28.2 ± 8.9 (5/5)
F-MNIST / Deep	n=500	58.2 ± 8.7 (5/5)	2.0 ± 0.0 (5/5)	11.0 ± 0.0 (5/5)	34.2 ± 7.4 (4/5)	21.0 ± 2.6 (5/5)	19.6 ± 1.9 (5/5)
F-MNIST / MLP	n=500	34.6 ± 1.9 (5/5)	11.8 ± 1.9 (5/5)	11.0 ± 0.0 (5/5)	— (0/5)	42.2 ± 12.6 (5/5)	33.0 ± 9.6 (5/5)
F-MNIST / Shallow	n=200	— (0/5)	16.6 ± 12.6 (5/5)	11.0 ± 0.0 (5/5)	— (0/5)	25.8 ± 10.3 (5/5)	20.6 ± 5.1 (5/5)
F-MNIST / Shallow	n=1000	73.0 (1/5)	38.0 ± 5.5 (5/5)	11.0 ± 0.0 (5/5)	— (0/5)	21.4 ± 4.0 (5/5)	21.2 ± 3.0 (5/5)
F-MNIST / Shallow	n=2000	71.0 ± 1.0 (2/5)	40.2 ± 4.0 (5/5)	11.0 ± 0.0 (5/5)	— (0/5)	18.4 ± 1.6 (5/5)	19.6 ± 3.9 (5/5)

Main observations.

1. Classical early stopping is the latest detector. It triggers in only 14 of 40 runs (35% rate), and when it does, it occurs on average at epoch 58–77, i.e., in the last quarter of training.

2. Weight norm triggers systematically at epoch 11. This very early signal is however non-specific: $\|\theta\|_2$ grows as soon as the optimizer updates parameters, independently of overfitting. This signal does not distinguish learning from memorization.

3. Loss gap (LG) exhibits high variance (σ up to 21.4 epochs on F-MNIST/Shallow), making it unreliable as a precise criterion.

4. Gradient norm (GN) only triggers on 2 of 8 configurations (CIFAR-10 and F-MNIST/Deep, 9/40 runs total), where gradient dynamics exhibit sharp spikes. On the remaining configurations, gradient norms increase too smoothly to exceed the 2× rolling mean threshold.

5. TDA detectors (smoothed derivative) achieve an interesting trade-off: they trigger on average around epochs 19–28 with a 100% detection rate on all 8 configurations. Inter-run variance is moderate ($\sigma \approx$ 2–10 epochs).

6. Subset size effect: on F-MNIST/Shallow, TDA-D triggers earlier as n_{train} increases (25.8 at $n = 200$, 18.4 at $n = 2000$), suggesting that larger training sets produce a faster and more pronounced topological transition.

7. Notable exception: MLP. On this architecture, TDA detectors are later (42.2 ± 12.6) than early stopping (34.6). We discuss this point in section 5.2.

4.3 Three-Phase Pattern

Across 6 of the 7 CNN configurations tested (the exception being F-MNIST/Deep), topological invariants follow a reproducible three-phase pattern:

Phase A — Compression (epochs 1–10). β_1 drops (e.g., 245 → 204 on F-MNIST/Shallow, seed 42) and PE(H_1) decreases (5.10 → 4.85). The network forms clusters and eliminates topological noise.

Phase B — Stabilization (epochs 10–40). β_1 and PE(H_1) stabilize in a plateau. This is the zone of relatively best generalization.

Phase C — Fragmentation (epochs 40–80). β_1 rises moderately (e.g., 223 $\rightarrow$ 227 on F-MNIST/Shallow, seed 42; mean +11.6 over 5 runs) and PE(H_1) re-increases (4.99 $\rightarrow$ 5.02; mean +0.077), signaling the gradual creation of new cyclic structures in the activation space. This rise coincides with the acceleration of the train-val gap.

4.4 Stability of Topological Invariants

Table 3 presents the bootstrap analysis results on the reference configuration (F-MNIST / CNN Shallow / $n = 500$), averaged over the 5 runs.

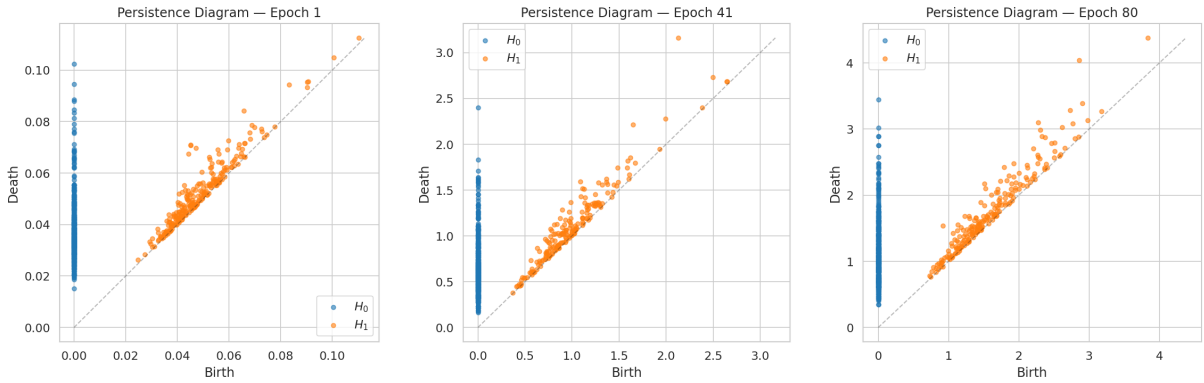
Table 3: Bootstrap analysis (10 draws at 80%, averaged over 5 runs). σ_{bs} = intra-bootstrap variance, σ_{run} = inter-run variance.

Epoch	PE(H_1)	σ_{bs}	σ_{run}	Ratio
1	4.741	0.047	0.079	0.59
10	4.529	0.052	0.055	0.94
20	4.587	0.048	0.086	0.56
40	4.586	0.052	0.078	0.66
60	4.609	0.058	0.056	1.04
80	4.685	0.046	0.039	1.18

The ratio $\sigma_{\text{bs}}/\sigma_{\text{run}}$ lies between 0.56 and 1.18, indicating that the variance introduced by 80% subsampling is of the same order of magnitude as the intrinsic training stochasticity. Topological invariants are therefore reasonably stable with respect to random sampling, but this stability degrades slightly at late epochs (ratio > 1 at epochs 60–80), precisely when overfitting fragments the activation space.

4.5 Persistence Diagrams

Figure 1 shows the persistence diagrams of the bottleneck activations at three representative epochs on the reference configuration (F-MNIST / CNN Shallow / $n = 500$). Each point (b, d) represents a topological feature born at scale b and disappearing at scale d; its distance to the diagonal $\ell = d - b$ measures its lifetime. Blue points correspond to H_0 features (connected components) and orange points to H_1 features (loops). At epoch 1, the diagram is dense, with many short-lived H_1 features close to the diagonal, reflecting a largely unstructured representation. By epoch 41, the number of H_1 points has decreased (245 to 224), while surviving features display significantly longer lifetimes (mean $\ell = 0.12$ vs. 0.004 at epoch 1), indicating that the network's internal representations have acquired a more structured topology. At epoch 80, H_1 features spread further from the diagonal (mean $\ell = 0.18$, max $\ell = 1.05$), with a comparable number of cycles (227) but increasingly persistent, reflecting a growing complexity in the activation space.



(a) Epoch 1 (b) Epoch 41 (c) Epoch 80

Figure 1: Persistence diagrams of the bottleneck at epochs 1, 41, and 80 (reference configuration: F-MNIST / CNN Shallow / $n = 500$).

4.6 Interpretation

The results in table 2 show that topological detectors (TDA-D) achieve a 100% detection rate on all 8 tested configurations, with a detection epoch significantly earlier than early stopping and comparable to or earlier than loss gap. However, it would be premature to conclude that topological invariants detect overfitting itself rather than a regime change in representation dynamics that correlates with overfitting.

The three-phase pattern (compression → stabilization → fragmentation) is consistent with results from Naitzat et al. [1] showing that successive layers simplify topology. Our contribution reveals that this temporal simplification (across epochs) is not monotonic: it partially reverses when the network begins to memorize.

Between epochs 40 and 80, the persistence diagrams in fig. 1 confirm this evolution: H_1 points (orange) progressively spread further from the diagonal, signaling the creation of new loops in the activation space that coincide with the acceleration of the train-val gap.

The rise in β_1 suggests that the network creates "protrusions" in the representation space to isolate memorized examples, generating spurious cyclic structures. This interpretation is consistent with work by Guss and Salakhutdinov [8] and Ramamurthy et al. [17].

5 Discussion

5.1 Comparison with Existing Approaches

The classical baselines fall into two categories. On one hand, weight norm and loss gap trigger early but lack specificity: the former reflects optimizer dynamics rather than overfitting, while the latter exhibits high variance (σ up to 21.4 epochs) that limits its reliability as an automatic criterion. On the other hand, early stopping is highly specific but too conservative: it requires val_loss to stagnate for 10 consecutive epochs, a condition that is rarely met in our setting (35% detection rate).

TDA-based detectors occupy a middle ground. TDA-D achieves a 100% detection rate with moderate variance, offering a signal that classical methods do not provide: a characterization of the geometry of internal representations, rather than scalar training statistics. This geometric perspective makes the detection complementary to, rather than redundant with, loss-based criteria. TDA-CUSUM is generally later but potentially more robust to local fluctuations in the topological time series.

5.2 Limitations

We identify several important limitations:

1. MLP case. TDA detectors are later than early stopping on MLP (42.2 vs. 34.6). This can be explained by the fact that MLPs overfit differently (no weight sharing), producing a less structured topological evolution. This result shows that the topological signal is not universally early.

2. Computational complexity. Vietoris-Rips computation in $\mathcal{O}(N^3)$ [10] imposes $N \leqslant 500$–1000 points, which constitutes severe subsampling for large datasets.

3. Layer choice. Only the bottleneck is analyzed. Other layers could present different or complementary signals.

4. Threshold calibration. CUSUM thresholds and smoothed derivative parameters (window size, warm-up period) are fixed a priori. An automatic calibration procedure would strengthen the method.

5. Experiment scale. Our architectures remain modest ($< 10^5$ parameters). Behavior on modern deep architectures (ResNets, Transformers) remains to be studied.

6. Potential non-specificity. On CNN configurations, TDA-D triggers around epochs 19–28 while the validation loss minimum occurs much later (epochs 64–80), meaning that the topological transition precedes actual generalization degradation by 40–60 epochs. The detected signal may therefore reflect a learning regime change without direct causal link to overfitting. Experiments with regularization (dropout, weight decay) would verify whether the signal disappears when overfitting is prevented.

5.3 Perspectives

- Integration of TDA signal into an adaptive early stopping criterion, combining $\mathrm{PE}(H_1)$ and val_loss, in order to benefit from both the earliness of the topological signal and the specificity of validation loss.

- Topological regularization: add a term penalizing the rise of β_1 in the loss function, in the spirit of neural persistence [7], so as to actively prevent the fragmentation phase rather than merely detect it.

- Extension to modern architectures (ResNet, Vision Transformer) and large-scale data (ImageNet), requiring persistent homology approximations (sparse filtrations, landmarks), to

assess whether the three-phase pattern generalizes beyond the small-scale architectures studied here.

- Study of higher-dimensional homology (H_2) to detect potential cavities in representations.

6 Conclusion

We conducted a systematic study of persistent homology of deep neural network activations as a tool for early detection of overfitting. Across 8 experimental configurations generated by a one-factor-at-a-time design over 3 datasets, 3 architectures, and 4 training sizes—with 5 independent runs each, we showed that:

1. Topological invariants (β_1, PE(H_1)) follow a three-phase pattern—compression, stabilization, fragmentation—reproducible across all tested CNN configurations.
2. Detectors based on the smoothed derivative of PE(H_1) and β_1 achieve a 100% detection rate on all 8 configurations, triggering at epochs 18–28 on CNN architectures—30 to 55 epochs before early stopping when it fires (35% of runs only).
3. Invariant stability with respect to subsampling is comparable to inter-run variability (bootstrap ratio ≈ 1).
4. The topological signal is however not universally early: it is later on MLP, calling for caution regarding generalization.

These results constitute an encouraging but not definitive proof of concept. Extension to larger-scale architectures and tasks, as well as verification of signal specificity (disappearance under regularization), are necessary steps before claiming that TDA offers a significant practical advantage for overfitting detection.

References

[1] Gregory Naitzat, Andrey Zhitnikov, and Lek-Heng Lim. *Topology of deep neural networks.* 2020. arXiv: `2004.06093 [cs.LG]`.

[2] Pierre Foret et al. *Sharpness-Aware Minimization for Efficiently Improving Generalization.* 2021. arXiv: `2010.01412 [cs.LG]`.

[3] Gunnar E. Carlsson. "Topology and data." In: *Bulletin of the American Mathematical Society* 46 (2009), pp. 255–308. DOI: `10.1090/S0273-0979-09-01249-X`.

[4] Frédéric Chazal and Bertrand Michel. *An introduction to Topological Data Analysis: fundamental and practical aspects for data scientists.* 2021. arXiv: `1710.04019 [math.ST]`.

[5] Edelsbrunner, Letscher, and Zomorodian. "Topological Persistence and Simplification." In: *Discrete & Computational Geometry* 28.4 (2002), pp. 511–533. DOI: `10.1007/s00454-002-2885-2`.

[6] Robert Ghrist. "Barcodes: The persistent topology of data." In: *Bulletin of the American Mathematical Society* 45 (2007), pp. 61–75. DOI: `10.1090/S0273-0979-07-01191-3`.

[7] Bastian Rieck et al. "Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology." In: *International Conference on Learning Representations*. 2019. URL: https://openreview.net/forum?id=ByxkijC5FQ.

[8] William H. Guss and Ruslan Salakhutdinov. *On Characterizing the Capacity of Neural Networks using Algebraic Topology*. 2018. arXiv: 1802.04443 [cs.LG].

[9] Ciprian Corneanu et al. *Computing the Testing Error without a Testing Set*. 2020. arXiv: 2005.00450 [cs.CV].

[10] Herbert Edelsbrunner and John Harer. *Computational Topology: An Introduction*. 2009. DOI: 10.1090/mbk/069.

[11] Nieves Atienza, Rocio Gonzalez-Díaz, and Manuel Soriano-Trigueros. "On the stability of persistent entropy and new summary functions for topological data analysis." In: *Pattern Recognition* 107 (2020). DOI: 10.1016/j.patcog.2020.107509.

[12] Han Xiao, Kashif Rasul, and Roland Vollgraf. *Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms*. 2017. arXiv: 1708.07747 [cs.LG].

[13] Y. Lecun et al. "Gradient-based learning applied to document recognition." In: *Proceedings of the IEEE* 86.11 (1998), pp. 2278–2324. DOI: 10.1109/5.726791.

[14] Alex Krizhevsky. *Learning Multiple Layers of Features from Tiny Images*. Tech. rep. 2009.

[15] Diederik P. Kingma and Jimmy Ba. *Adam: A Method for Stochastic Optimization*. 2017. arXiv: 1412.6980 [cs.LG].

[16] Ulrich Bauer. "Ripser: efficient computation of Vietoris–Rips persistence barcodes." In: *Journal of Applied and Computational Topology* 5.3 (2021), pp. 391–423. DOI: 10.1007/s41468-021-00071-5.

[17] Karthikeyan Natesan Ramamurthy, Kush R. Varshney, and Krishnan Mody. *Topological Data Analysis of Decision Boundaries with Application to Model Selection*. 2018. arXiv: 1805.09949 [stat.ML].